balanced_leverage_max_legacy

equity strategy | family: equity-max | status: reference | derived from `balanced_leverage_max`

What this strategy does

balanced_leverage with bigger bets for more return: base unit 0.012, 6x leverage cap -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Market-neutral earnings drift with enough gross exposure to matter; the neutrality is the hedge.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.012
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	balanced leverage max legacy	market (CRSP VW)
Annualized return	12.15%	9.94%
Annualized volatility	12.20%	19.35%
Sharpe (NAV-pct)	1.00	0.59
Sortino	1.45	
Max drawdown	-36.5%	-55.5%
Calmar	0.33	
Final NAV ($10M start)	$93,270,642	
Worst day	-7.14%	
% positive days	51.5%	

Sharpe 1996-2001	1.53	
Sharpe 2002-2007	1.71	
Sharpe 2008-2013	-0.07	

Correlation of daily returns with the market: **-0.02**.

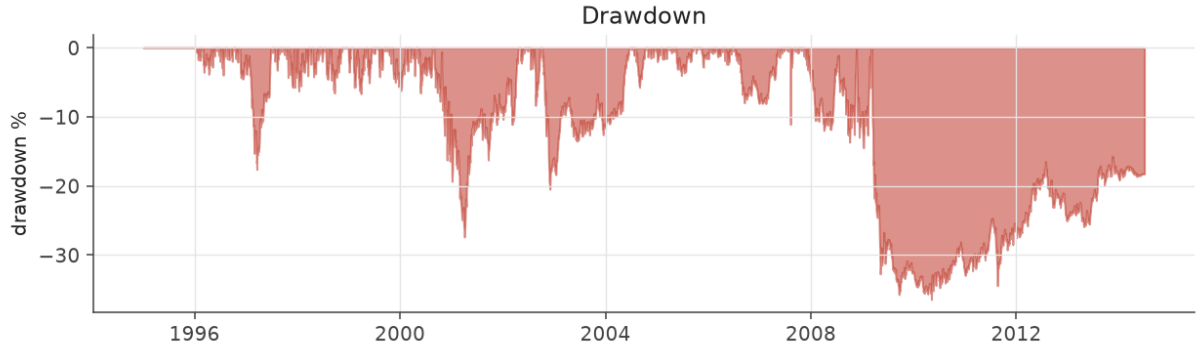
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1

• best return among equity: `market_boost_max` (0.116); this one: 0.121

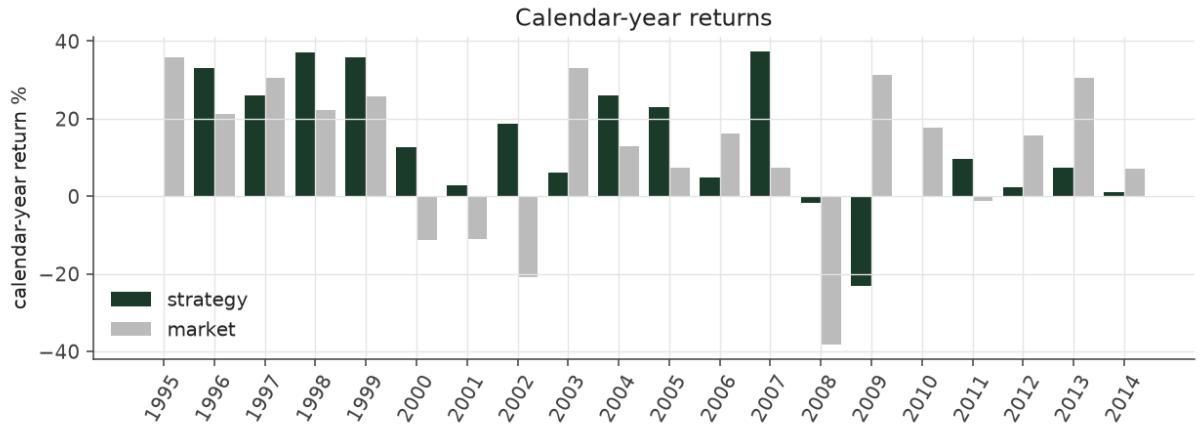
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.333



NAV (log scale)



Drawdown from running peak

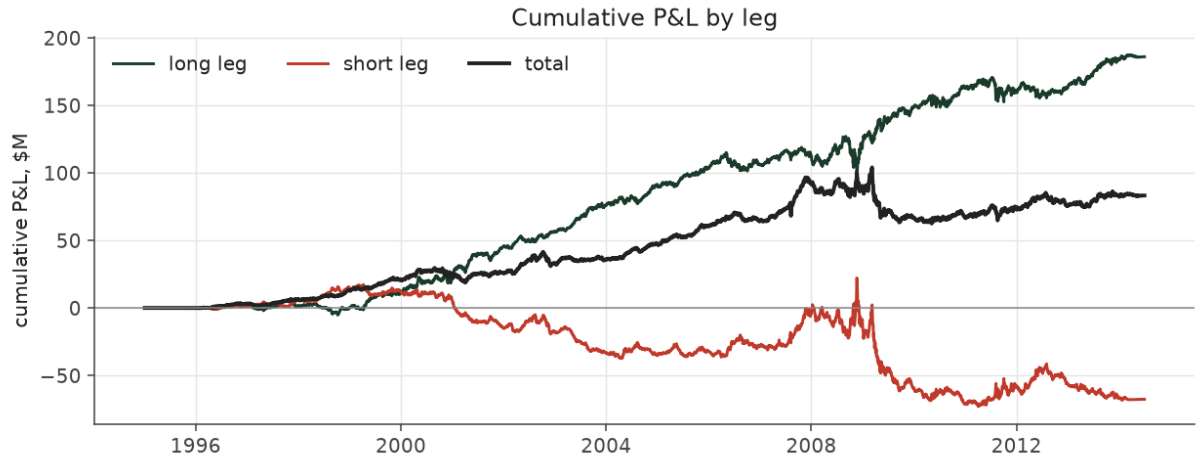


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	185.98	-67.62
P&L after costs ($M)	168.87	-85.60
of which common (average-event) component ($M)	175.41	-173.00

of which excess / PEAD component ($M)	9.52	105.97
excess minus costs ($M)	-7.56	88.03
gross return per $ traded (bp)	166.77	-59.88
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.06	-0.22

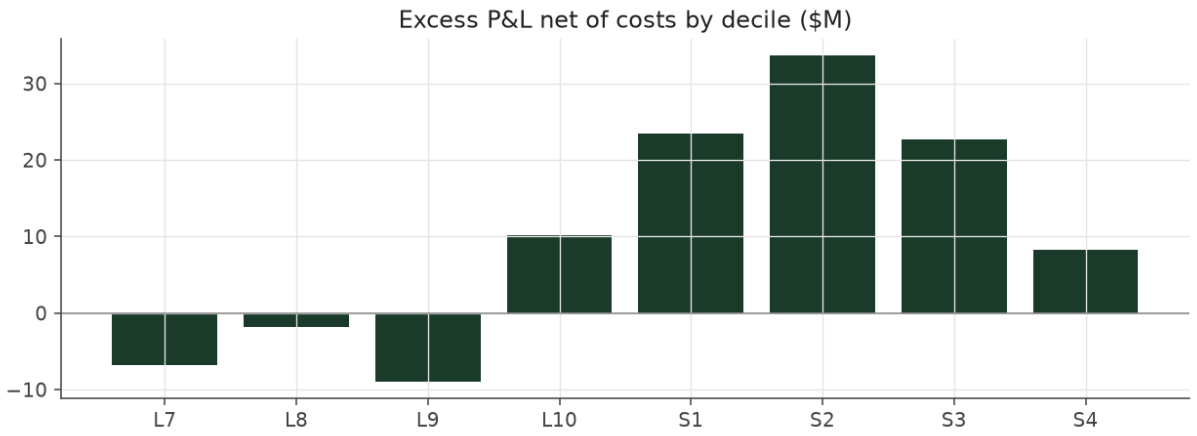
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-67.6M before costs) while the long leg earns +186.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 175.4M / 173.0M. Stripping that common component, the surprise signal itself earned +9.5M on the long side and +106.0M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.06 long / -0.22 short vs 1.00 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	7,718	1,052.78	11.28	-5.18	1.67	-6.85	-8.5%	-65.07
long	8	16,225	2,721.85	42.43	2.34	4.23	-1.89	-2.3%	-6.95
long	9	17,480	3,520.62	43.56	-3.69	5.32	-9.01	-11.2%	-25.59
long	10	16,939	3,857.22	71.60	16.06	5.90	10.16	12.6%	26.33
short	1	16,755	3,662.36	-27.10	29.32	5.92	23.40	29.1%	63.89
short	2	16,916	3,529.55	-24.03	39.18	5.54	33.64	41.8%	95.32
short	3	15,579	2,858.54	-23.05	27.22	4.57	22.65	28.1%	79.22
short	4	9,015	1,241.23	-11.41	10.25	1.95	8.30	10.3%	66.91



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,943	580.43	4.74	8.80	1.91	6.89	8.6%	118.74
2	22,307	2,425.20	15.93	23.41	5.94	17.47	21.7%	72.03

3	25,252	4,605.79	25.85	32.96	8.66	24.31	30.2%	52.78
4	27,425	6,685.72	17.20	24.36	9.76	14.60	18.1%	21.84
5	28,700	8,147.02	19.55	25.97	8.84	17.13	21.3%	21.03

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	11,934	272.18	5.63	7.61	1.36	6.25	7.8%	229.46
2	22,802	1,937.75	12.15	20.53	5.81	14.71	18.3%	75.93
3	25,740	4,782.62	26.45	37.55	9.56	27.99	34.8%	58.52
4	27,494	7,253.58	21.14	28.67	10.15	18.51	23.0%	25.52
5	28,652	8,196.33	17.81	21.03	8.20	12.83	15.9%	15.65

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,165	9,600.44	-14.15	35.15	14.06	21.09	26.2%	21.97
41-60d	25,072	4,938.28	27.47	18.39	7.84	10.56	13.1%	21.38
61-90d	34,592	6,224.84	23.02	26.24	10.15	16.09	20.0%	25.84
>90d	11,798	1,680.60	46.93	35.71	3.04	32.67	40.6%	194.40

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,056	3,008.84	26.65	30.08	4.72	25.36	31.5%	84.28
2002-07	39,668	7,889.54	61.23	69.53	12.19	57.34	71.3%	72.68
2008-13	44,903	11,545.77	-4.60	15.88	18.18	-2.30	-2.9%	-1.99

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,429	2,583.01	6.03	8.57	4.33	4.24	5.3%	16.40
Chems	2,571	1,235.93	7.33	8.92	1.67	7.26	9.0%	58.73
Durbl	2,800	1,290.38	2.69	4.58	1.78	2.81	3.5%	21.73
Enrgy	6,106	1,767.37	16.94	20.27	2.50	17.77	22.1%	100.55
Hlth	11,413	2,280.39	12.88	14.71	3.68	11.03	13.7%	48.37
Manuf	11,825	2,139.00	5.43	8.60	3.40	5.20	6.5%	24.31
Money	17,141	2,149.44	11.87	16.66	3.75	12.91	16.0%	60.07
NoDur	5,000	1,595.65	5.49	6.46	2.31	4.15	5.2%	26.03
Other	18,022	2,447.24	8.16	13.06	4.30	8.76	10.9%	35.78
Shops	11,767	2,235.81	13.10	15.48	3.56	11.93	14.8%	53.34

Telcm	3,932	1,386.74	-7.51	-4.84	1.87	-6.71	-8.3%	-48.41
Utils	4,621	1,333.20	0.86	3.03	1.96	1.06	1.3%	7.98

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	38,814	8,085.79	2.77	19.33	11.77	7.57	9.4%	9.36
low vol	39,075	6,848.95	38.30	46.76	11.52	35.24	43.8%	51.45
mid vol	37,173	7,280.81	41.91	49.66	11.41	38.26	47.5%	52.54

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	12.15%	-0.00%
0.5%	10.7	11.45%	0.70%
1.0%	21.5	10.65%	1.50%
2.0%	43.0	8.65%	3.50%
5.0%	107.4	-100.00%	112.15%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.